School Worksheet

Class 12 Mathematics, Chapter 4: Determinants

Maximum marks: **54** **20** questions

Name _____ Date _____ Score _____ / 54

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ — 1 mark

If $\begin{vmatrix} x & 2 \\ 18 & x \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 18 & 6 \end{vmatrix}$, then x is equal to:

(A) 6 **(B)** ± 6

(C) -6 **(D)** 0

2 MCQ — 1 mark

If A is a square matrix of order 3 with $|A| = 5$, then $|\text{adj } A|$ is:

(A) 5

(B) 25

(C) 125

(D) $\frac{1}{5}$

3 MCQ — 1 mark

If A is a square matrix of order 3 and $|A| = 4$, then $|2A|$ is:

(A) 8 **(B)** 16

(C) 32 **(D)** 64

4 MCQ — 1 mark

The area of the triangle with vertices $(-2, -3)$, $(3, 2)$ and $(-1, -8)$ is:

(A) 15 sq units **(B)** 30 sq units

(C) 20 sq units **(D)** 10 sq units

5 MCQ — 1 mark

The cofactor of the element a_{12} in $\begin{vmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{vmatrix}$ is:

(A) 46 **(B)** -46

(C) 4 **(D)** -4

Q No.	Question	Marks
6	Assertion-Reason **Assertion (A):** If A is a 3×3 matrix with $\lvert A\rvert = 2$, then $\lvert A\,(\text{adj}\ A)\rvert = 8$. **Reason (R):** For any square matrix A, $A\,(\text{adj}\ A) = \lvert A\rvert\, I$. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer Find the values of k if the area of the triangle with vertices $(k, 0)$, $(4, 0)$ and $(0, 2)$ is 4 sq units.	2 marks

Level 2

17 marks

8	Short answer Find the adjoint and inverse of $A = \begin{bmatrix} 2 & -1 \\ 4 & 3 \end{bmatrix}$.	2 marks
9	Short answer If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, show that $A^2 - 5A + 7I = O$. Hence find A^{-1}.	3 marks
10	Short answer Using determinants, find the equation of the line joining the points $(1, 2)$ and $(3, 6)$.	2 marks
11	Short answer Solve the system of equations using the matrix method: $5x + 2y = 4$, $7x + 3y = 5$.	3 marks
12	Short answer Examine the consistency of the system: $x + 3y = 5$, $2x + 6y = 8$.	2 marks
13	Short answer If $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$, show that $A^{-1} = \frac{1}{19}A$.	2 marks
14	Short answer If $A = \begin{bmatrix} 2 & 3 \\ 1 & -4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$, verify that $(AB)^{-1} = B^{-1}A^{-1}$.	3 marks

Q No.	Question	Marks

Level 3 — 29 marks

15 Long answer — 5 marks

A school wants to allocate students into three clubs: Sports, Music and Drama, under the following conditions: (i) the number of students in the Sports club should be equal to the sum of the numbers of students in the Music and Drama clubs; (ii) the number of students in the Music club should be 20 more than half the number of students in the Sports club; (iii) the total number of students allocated to the three clubs is 180. Using the matrix method, find the number of students allocated to each club.

16 Long answer — 5 marks

Find the product $\begin{bmatrix}1 & -1 & 2\\ 0 & 2 & -3\\ 3 & -2 & 4\end{bmatrix}\begin{bmatrix}-2 & 0 & 1\\ 9 & 2 & -3\\ 6 & 1 & -2\end{bmatrix}$ and use it to solve: $x - y + 2z = 1$, $2y - 3z = 1$, $3x - 2y + 4z = 2$.

17 Long answer — 5 marks

If $A = \begin{bmatrix}1 & 2 & -3\\ 2 & 3 & 2\\ 3 & -3 & -4\end{bmatrix}$, find A^{-1}. Hence solve: $x + 2y - 3z = -4$, $2x + 3y + 2z = 2$, $3x - 3y - 4z = 11$.

18 Long answer — 5 marks

The cost of 4 kg onion, 3 kg wheat and 2 kg rice is ₹60. The cost of 2 kg onion, 4 kg wheat and 6 kg rice is ₹90. The cost of 6 kg onion, 2 kg wheat and 3 kg rice is ₹70. Find the cost of each item per kg by the matrix method.

19 Long answer — 5 marks

If $A = \begin{bmatrix}1 & 2 & 0\\ -2 & -1 & -2\\ 0 & -1 & 1\end{bmatrix}$, find A^{-1}. Using A^{-1}, solve: $x - 2y = 10$, $2x - y - z = 8$, $-2y + z = 7$.

20 Case study — 4 marks

Case Study: A farmer owns a triangular plot of land whose corners, on a map grid, are $A(2, 3)$, $B(-1, 0)$ and $C(3, -2)$ (1 unit = 10 m). He plans to fix a pole at a point $D(k, 1)$ on the boundary line AB.

(i) Using determinants, find the area of the plot in square units. — 1 mark

(ii) Find the value of k so that D lies on line AB. — 1 mark

(iii) Using determinants, find the equation of line AB. — 2 marks

Answer key

Chapter 4: Determinants, School Worksheet. Full step-wise solutions are in the solutions sheet.

Level 1

1	(B) ± 6
2	(B) 25
3	(C) 32
4	(A) 15 sq units
5	(A) 46
6	(A) Both A and R are true, and R is the correct explanation of A
7	$k = 0$ or $k = 8$

Level 2

8	$\operatorname{adj} A = \begin{bmatrix} 3 & 1 \\ -4 & 2 \end{bmatrix}$, $A^{-1} = \frac{1}{10}\begin{bmatrix} 3 & 1 \\ -4 & 2 \end{bmatrix}$
9	$A^2 - 5A + 7I = O$ shown; $A^{-1} = \frac{1}{7}\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$
10	$y = 2x$
11	$x = 2, y = -3$
12	Inconsistent (no solution), since $\lvert A\rvert = 0$ and $(\operatorname{adj} A)B \neq O$
13	$A^{-1} = \frac{1}{19}\begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix} = \frac{1}{19}A$ (shown)
14	$(AB)^{-1} = B^{-1}A^{-1} = \frac{1}{11}\begin{bmatrix} 14 & 5 \\ 5 & 1 \end{bmatrix}$ (verified)

Level 3

15	Sports $= 90$, Music $= 65$, Drama $= 25$ students
16	Product $= I$; $x = 0, y = 5, z = 3$
17	$A^{-1} = \frac{1}{67}\begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}$; $x = 3, y = -2, z = 1$
18	Onion ₹5 per kg, wheat ₹8 per kg, rice ₹8 per kg
19	$A^{-1} = \begin{bmatrix} -3 & -2 & -4 \\ 2 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}$; $x = 0, y = -5, z = -3$
20	(i) 9 sq units (ii) $k = 0$ (iii) $x - y + 1 = 0$